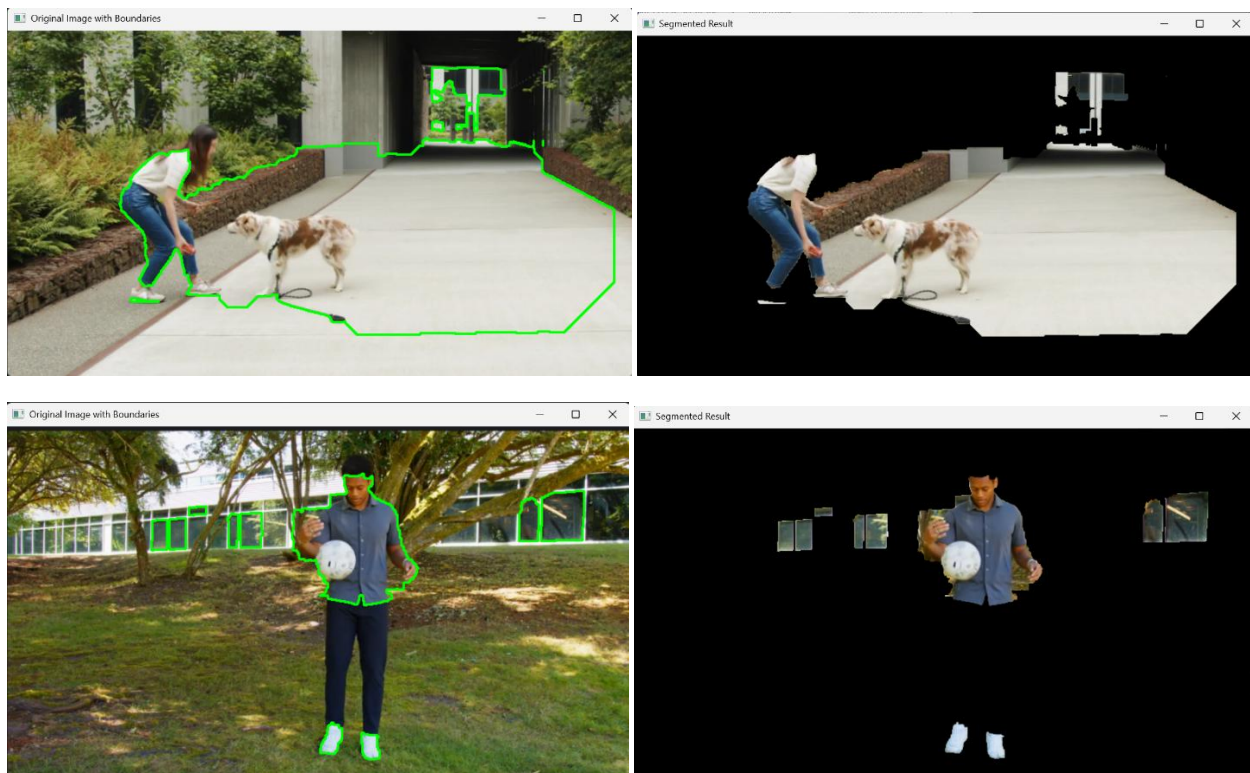


GitHub Repository: https://github.com/yosammy01/CSC8830_COMPUTER_VISION/tree/main/MODULE4

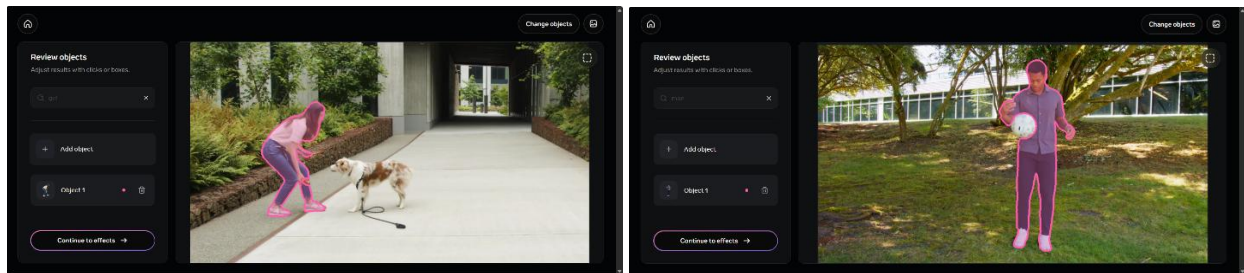
Web Application: https://yosammy01.github.io/CSC8830_COMPUTER_VISION/#/module4

- I. Implement: Implement a script to find the exact boundaries of a human in an image from a regular color (RGB) imaging camera.
 - a. Using the functions and libraries available in opencv, human_segmentation.py was created. Deep learning and machine learning based code was not used.
 - b. The script utilizes opencv's grabCut method, which is an implementation of the computer vision GrabCut algorithm for image segmenting.
 - c. Comparing results with SAM2 segmentation (<https://ai.meta.com/research/sam2/>)
 - i. The computer vision algorithm struggled to segment the human beings in the images.
 - ii. The SAM2 demo has a clear advantage in segmenting human beings.
 - iii. The machine learning model being used in the SAM2 demo knows what a human being is and can identify and isolate a human in an image from other objects in the background or foreground.

human_segmentation.py output:

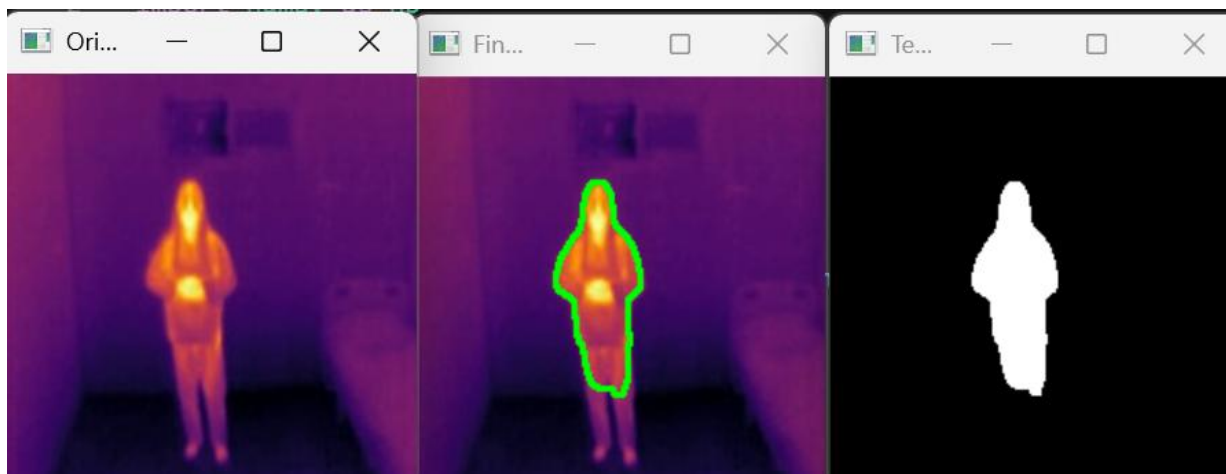


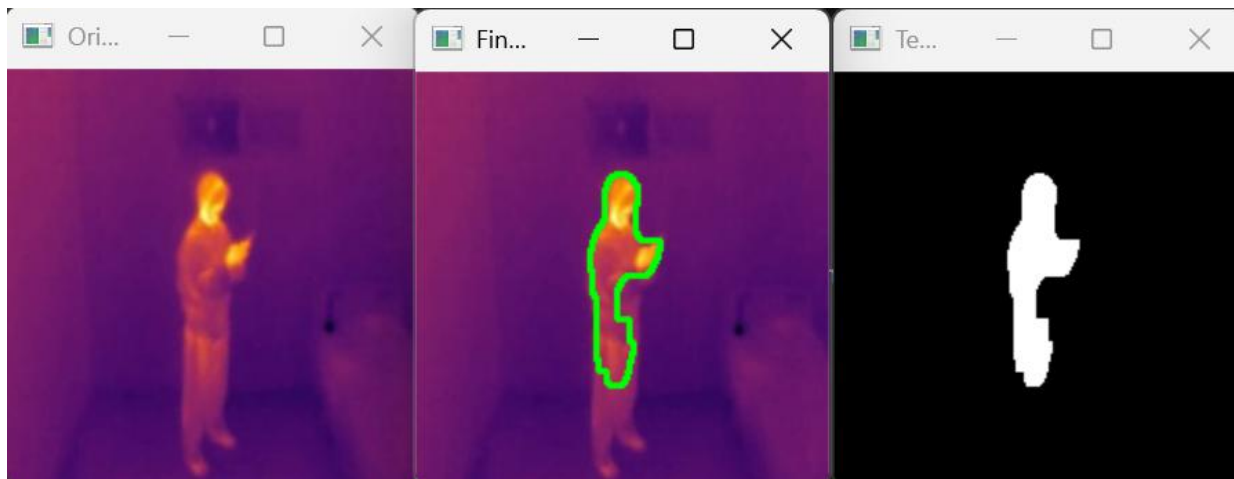
SAM2 Demo:



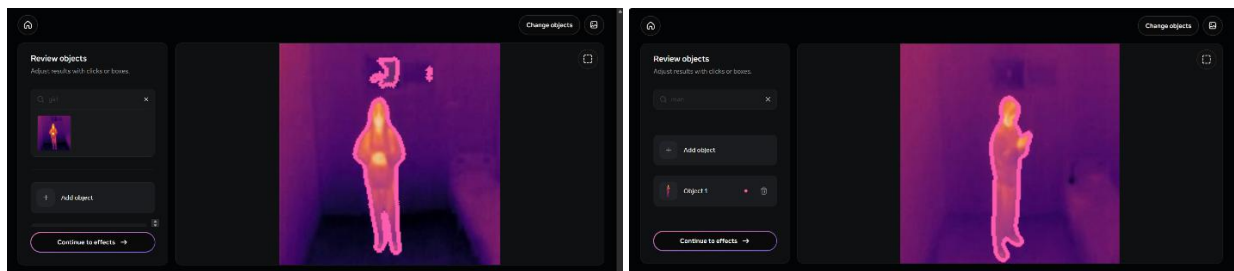
- II. Implement a script to find the exact boundaries of a human in an image from a thermal imaging camera (e.g. <https://blog.roboflow.com/thermal-infrared-dataset-computer-vision/>).
- Using the functions and libraries available in opencv, thermal_segmentation.py was created. Deep learning and machine learning based code was not used.
 - The script utilizes opencv's threshold and morphologyEx methods, which are implementations of the computer vision Otsu's Thresholding and Morphology algorithms.
 - Comparing results with SAM2 segmentation (<https://ai.meta.com/research/sam2/>)
 - The computer vision algorithms performed better for thermal images, than the RGB images. There were still some missed parts of the humans due to the clothing being cold.
 - The SAM2 demo has a clear advantage in segmenting human beings in thermal images.
 - The machine learning model being used in the SAM2 demo knows what a human being is and can identify and isolate a human in a thermal image from other objects in the background or foreground. There was one odd anomaly where a background object was picked up in the thermal image by the SAM2 demo, as part of the girl.

Thermal segmentation.py output:





SAM2 Demo:



- III. Theory:
- Explanation with mathematical derivations on how edge detection and segmentation of regions can be achieved using Fourier (frequency) domain analysis.

Samuel
Robles
CSC 8830
Module 4

Theory:

- ① We know that convolution of a function and a filter or kernel is equivalent to multiplying the Fourier transforms of the function and filter.
- ② From Computer Vision, it is known that the Fourier transform of the 2D spatial image ~~represents~~ represents the image in the frequency domain. Also, the low frequencies of the Fourier transformed image directly relate to the homogeneous regions (no or low intensity changes) of the spatial image. Likewise, the high frequencies directly relate to the edges or noise (high intensity changes).

③ Edge Detection:

The Laplacian (second-order derivative) of the spatial function can be used to find edges with "zero-crossings". The Laplacian can be Fourier-transformed with the following formula:

$$\text{Laplacian: } \nabla^2 f(x, y) = f''(x) + f''(y)$$

Fourier Transform of Laplacian:

$$\mathcal{F}\{\nabla^2 f(x, y)\} = -4\pi(u^2 + v^2)F(u, v)$$

$$\text{Note: } F(u, v) = \mathcal{F}\{f(x, y)\}$$

So, the edges of an image can be found by applying a high-pass filter to remove the low frequencies of the image. This can be done in the frequency domain by multiplying the Fourier transform of the image with the Fourier transform of the high-pass filter.

$$\mathcal{F}\{\nabla^2 f(x, y) * h(x, y)\} =$$

$$\mathcal{F}\{f''(x) + f''(y)\} \cdot H(u, v)$$

$$[-4\pi^2(u^2 + v^2)F(u, v)] \cdot H(u, v)$$

④ Region Segmentation:

Similar to edge detection, the regions of an image to be segmented can be found by applying a low-pass filter to remove the high frequencies of the image. This can be done in the frequency domain by multiplying the Fourier transform of the image with the Fourier transform of the low-pass filter.

$$\mathcal{F}\{f(x, y) * h(x, y)\} = F(u, v) \cdot H(u, v)$$